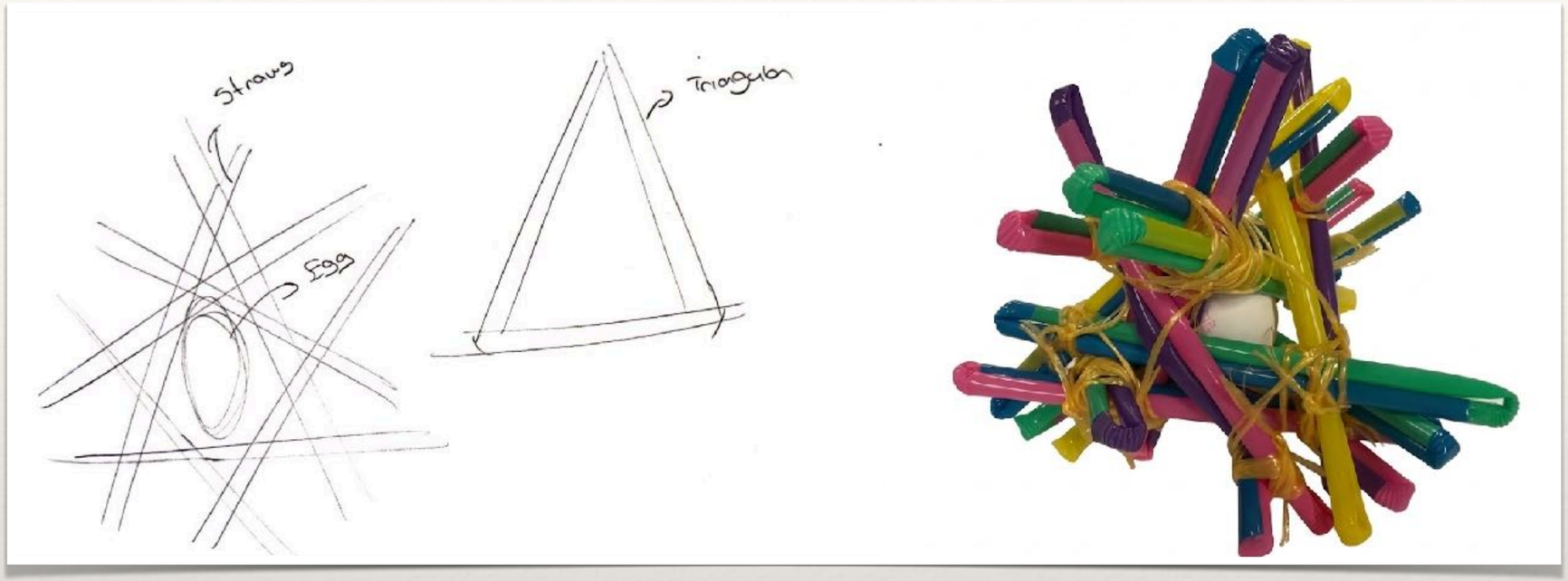
IDEXAI

StrawGuard

Çınarhan Çatal



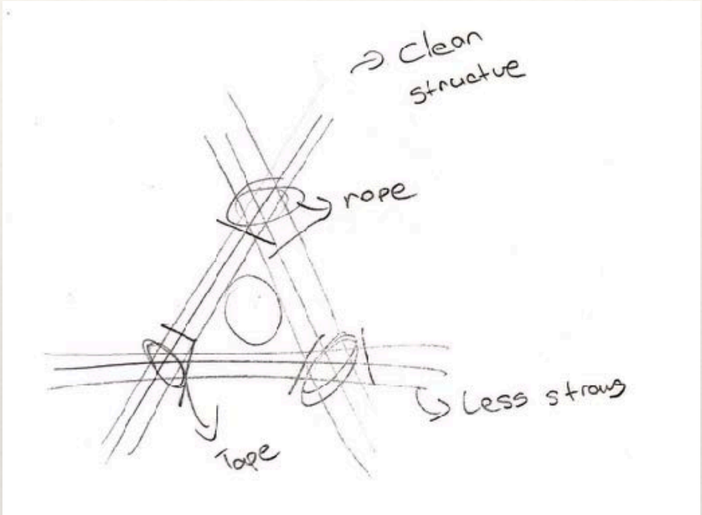
Design Challenge

The challenge was to design a structure that protects a raw egg during a drop test while using simple lightweight materials

Interaction with AI

Clarifier Prompt:

What are the key factors that cause an egg to break during a drop?

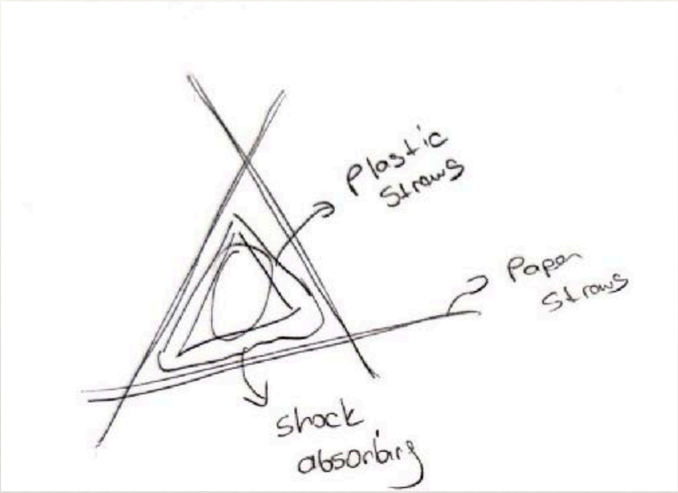


AI Response:

Impact force
Direct pressure
Lack of structural support

Expander Prompt:

What lightweight structures could absorb impact during a drop test?



AI Response Summary:

frame structures
suspension systems
shock absorbing shapes

Final Production

Drop test:

1.5m survived ✓

3.0m survived ✓

Final Reflection

This project helped me understand how structural design can influence impact protection. Through experimentation and reflection, I learned that distributing forces and creating stable structures are essential for protecting fragile objects.